Towards Subseasonal-to-Seasonal Drought-to-Flood Prediction

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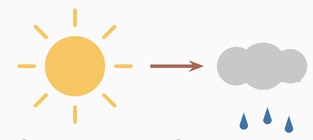
Introduction

Floods and droughts can occur in sequence — a **drought-to-flood transition (D2F)**. Globally, ∼**25% of floods are preceded by drought** [2], and socio-economic losses can be up to **8x greater** than from isolated hazards [3]. Forecasting D2Fs at subseasonal-to-seasonal lead times could support early action, but 2 research gaps remain ⟶

No catalogue of D2F events exists for Ireland

↓ Current focus

Subseasonal-to-seasonal **forecast skill** over Ireland has **not been assessed**



Subseasonal-to-seasonal D2F prediction

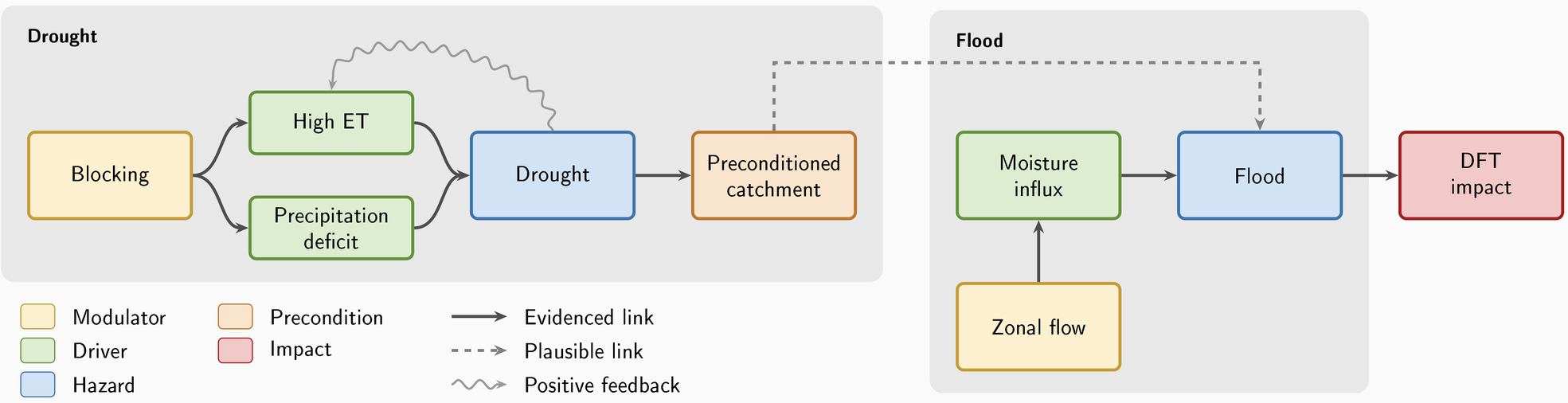


Fig. 1: Physical pathway for D2F events, after the framework of Bevacqua et al. [1].

Methods

As a first step towards D2F prediction, we verify **2 m temperature** forecasts from the ECMWF subseasonal ensemble at **weeks 2–6** over Ireland on the native O320 grid, stratified by season. To remove systematic biases, forecasts and ERA5 observations are each referenced to their own climatology (±15-day window, 1991–2020); the ERA5 climatology is computed on its N320 grid and bilinearly interpolated to O320. Skill is assessed with **centred ACC** and **RMSE**; ensemble calibration with **rank histograms** [4].

Data

ECMWF ENS sub-seasonal (CY49R1)

11 members ▪ init. every 2 days ▪ weeks 2–6

2 m temperature ▪ O320 grid ▪ 2006–2016

Reference data

ERA5 reanalysis ▪ Met Éireann observations

Results

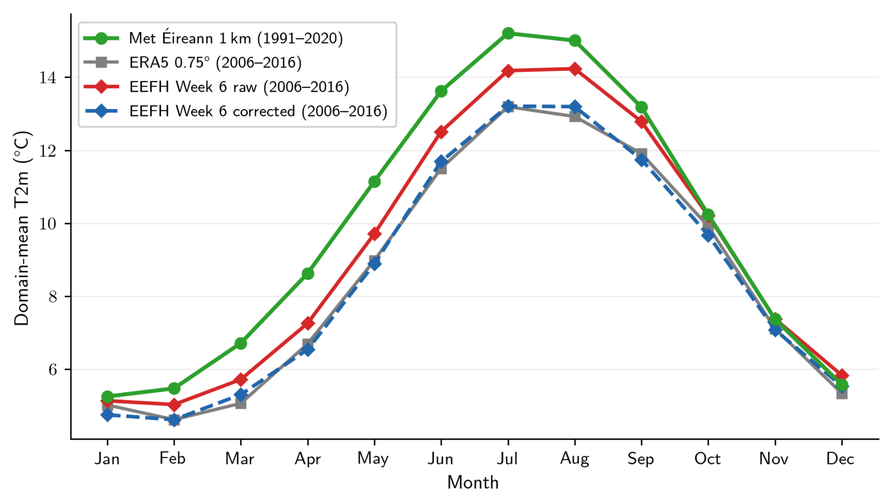


Fig. 2: Domain-mean 2 m temperature: forecast (weeks 2–6), ERA5, and Met Éireann observations.

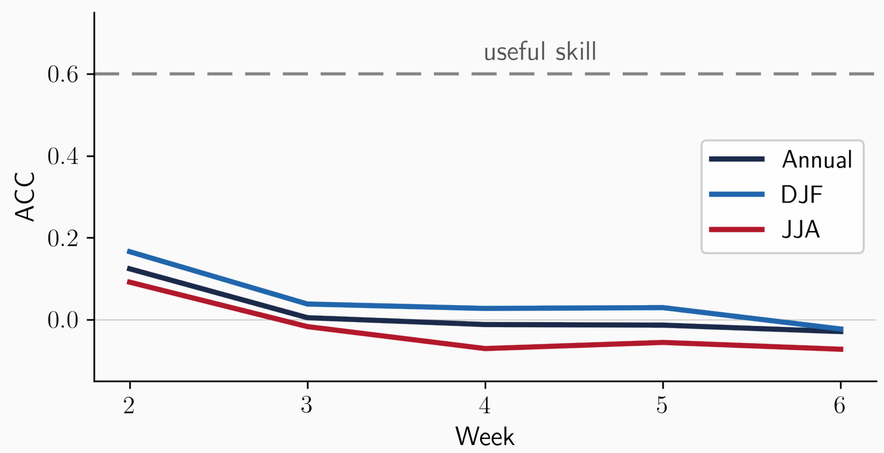


Fig. 3: (a) Centred ACC and (b) anomaly bias by lead time and season.

RMSE MAPS
(scoring in progress)

Fig. 4: Anomaly RMSE at weeks 2, 4, and 6 over Ireland (O320 grid).

- JJA warm bias is consistent with known IFS limitations in **soil moisture** and **vegetation–atmosphere coupling** [5]
- The forecast **outperforms climatology** to week 6; **DJF skill exceeds JJA** at early leads

Future Work

Compare with SEAS5 seasonal forecasts ⟶ Evaluate probabilistic forecast skill ⟶ Apply probabilistic bias correction

References

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[4] Wilks (2011). *Statistical Methods in the Atmospheric Sciences*, 3rd ed. [5] Hersbach et al. (2020). *Q. J. R. Meteorol. Soc.*, 146, 1999–2049.